Week3_Individual_Assignment_Sulaiman (Stan)_Abu_Romman

Section 1: Introduction

This individual task is part of Week 3's "Welcome to the World of Objects" assignment. The goal is to understand the concept of a domain model and to create a simple domain model for a selected business case. I chose the Online Food Ordering System, which enables customers to browse menus, place orders, and receive deliveries.

Section 2: What is a Domain Model?

A domain model, also referred to as a class diagram, is a visual representation of the main concepts, objects, and relationships in a system. It helps describe what exists in the system, such as customers, orders, and products, without focusing on how it is implemented in code. Each concept is represented as a class, which includes its attributes and associations with other classes.

Section 3: Why is the Domain Model used?

A domain model is used to help understand, analyze, and communicate the structure of a system before programming it. It provides a shared understanding between developers, designers, and stakeholders. It also assists in identifying what data will be needed, how different parts connect, and what relationships exist in the system. In short, it is a planning tool that makes later stages of software design easier, simpler, and more accurate.

Section 4: Typical Elements of a Domain Model

A domain model typically includes the following elements:

1. Classes – the main entities or objects in the system, e.g., Customer, Order, Restaurant.
2. Attributes – the pieces of information each class stores, e.g., name, email, or price.
3. Associations – relationships between classes, e.g., Customer places Order.

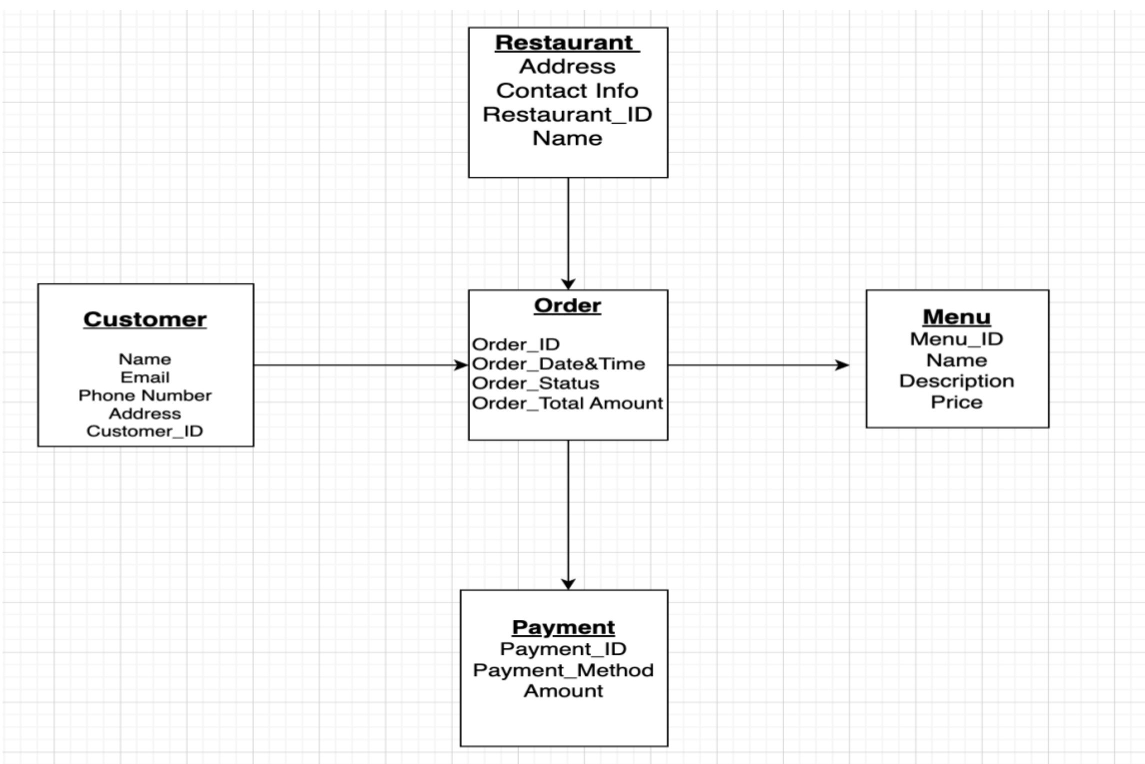
4. Multiplicities – how many objects are connected, e.g., one Customer can place many Orders.

Section 5: Example: Stomach Saver (Online Food Ordering System)

In my business case, the system is called Stomach Saver. It is an online food ordering platform that allows customers to browse restaurant menus, place food orders, and receive deliveries at their homes, workspaces, or anywhere they like. The idea behind Stomach Saver is to make ordering food quick, reliable, and easy for everyone.

The main classes in the Stomach Saver system are Customer, Restaurant, Item menu, Order, Order Item, Delivery Staff, and Payment. Each class represents a different part of how the system functions. Customers can browse menus and place orders. Restaurants prepare the food and update menu items. Delivery staff members deliver the orders to customers, and each order has a connected payment record.

Below is my first draft of a simple domain model for Stomach Saver, showing the relationships between these classes.



This model represents the key data structure of the system before moving to implementation.